

CRISPResso2

UNIVERSITY OF
WESTMINSTER



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4th year Ph.D Biotechnology,
Genome Engineering Laboratory,
University of Westminster, London.

Tuesday 29th July 2025

1992 – 2017 Freelance Software Engineer / Technical Team lead

Java / C/C++ / Perl / HTML/CSS, SQL, full stack developer, SCRUM Master. Clients: Economist.com, Royal Bank of Scotland, Credit Suisse, TAG Worldwide, Ocado, Royal Navy. Experiments with Blockchain/Crypto and FX.

2018 – 2022 BSc Biological Sciences

CRISPR/Cas9, **sickle cell disease** cure (in 2018 first human trials were being considered), editing mosquito DNA to change **eye colour**, Dr He Jianku twins controversy, introduced to Dr Kalpana and the Genome Engineering Laboratory)

2020 – 2021 The Gene Editors of The Future programme (first iteration)

Designed guides with a plan to **target multiple sites** in the Q-arm of chromosome 14. Covid prevented us from working in the laboratory during that period.

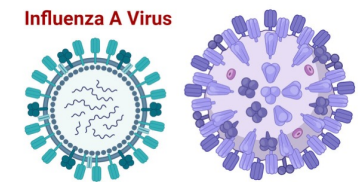
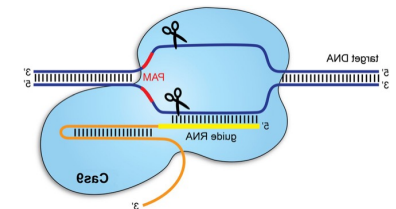
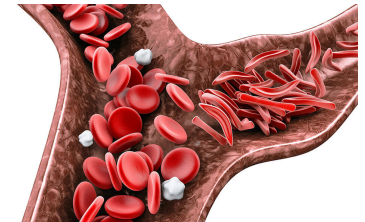
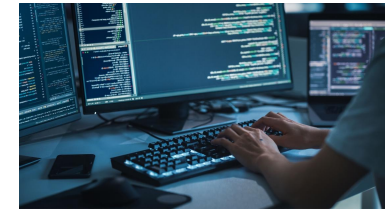
2021-2022 – final year project

Developed a CRISPR based tool for **detection and diagnosis** of Influenza A virus

2022 – present – PhD Biotechnology

Investigating the genomic and functional consequences of CRISPR/Cas9-mediated **ZFP36L1 knockout** in MCF-7 breast cancer cells. My research focuses on how loss of this RNA-binding protein (known for regulating mRNA stability) impacts replication stress, mutational burden, and genome integrity, particularly under treatment of replication stress inducing agents. Using whole-genome sequencing, bioinformatics, and cell culture models, I explore *ZFP36L1*'s role as a potential genome guardian and cancer vulnerability factor..

the journey so far ...



What is CRISPResso2?

- **CRISPResso2** is a bioinformatics tool for analysing CRISPR genome editing outcomes using sequencing data.
- Quantifies editing efficiency, including insertions, deletions (indels), and nucleotide substitutions.
- Provides high-resolution visualization of edits at the single-nucleotide level.
- Supports analysis of both single-locus and pooled amplicon experiments.
- Compatible with various CRISPR systems, including Cas9 and base editors.
- Widely used to assess the precision and effectiveness of gene editing in targeted genomic regions.

What questions can it help to answer?

Did my CRISPR experiment produce edits at the target site?

(i.e., was there any evidence of genome editing?)

What was the editing efficiency?

(Percentage of reads containing insertions, deletions, or substitutions)

What types of edits occurred?

(Breakdown of indels, substitutions, and their frequencies)

Were there frameshift vs. in-frame mutations?

(Useful for coding regions to infer functional consequences)

What is the distribution of editing outcomes across the cell population?

(Visualization of heterogeneity in editing)

Developers of CRISPResso2

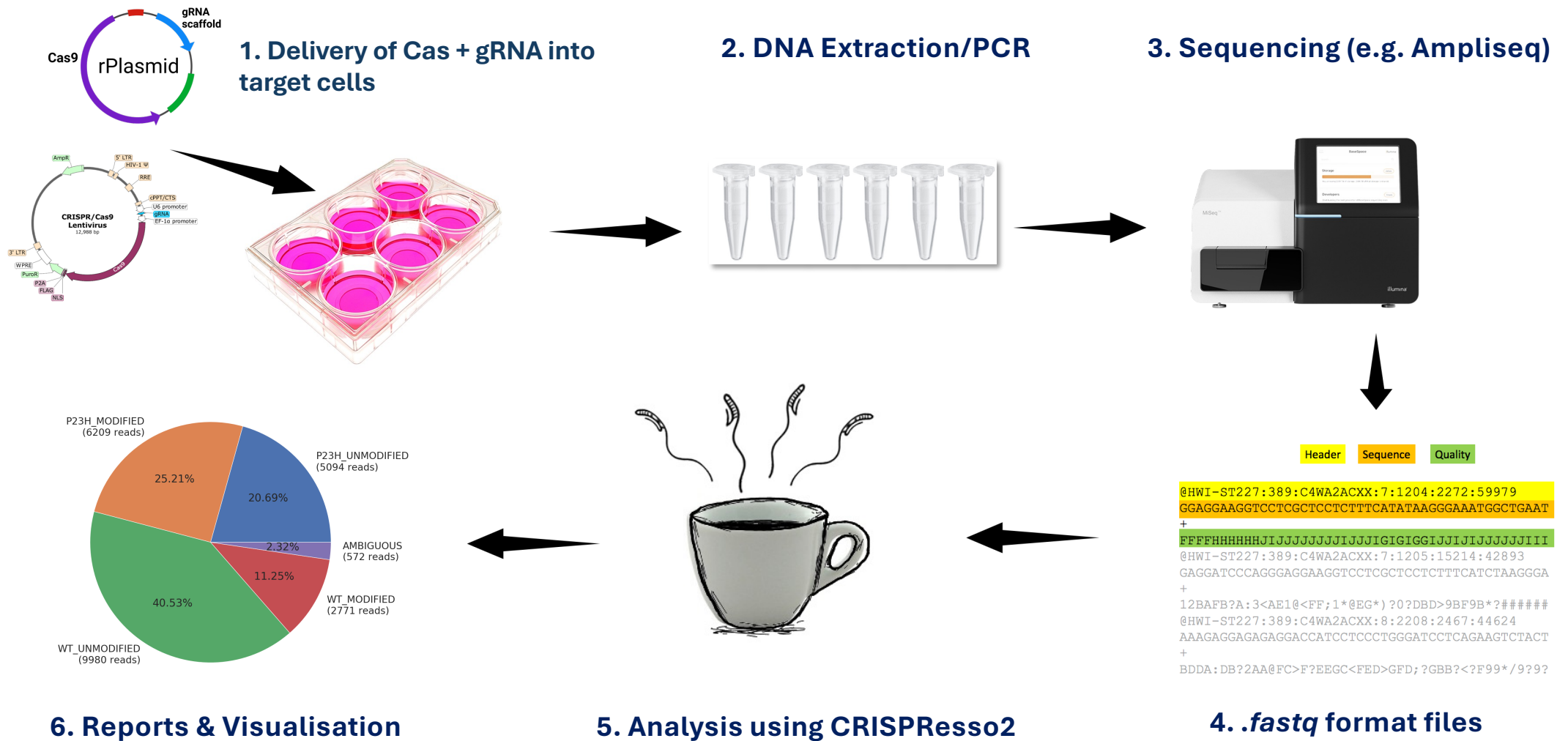


Luca Pinello
Ph.D., Associate Professor
Massachusetts General
Hospital and HMS
(CRISPResso)



Kendell Clement
Assistant Professor,
Biomedical Informatics
University of Utah,
(CRISPResso2)

CRISPResso2 in the Genome Editing Workflow



Format of *.fastq* file

Line 1: Header

@HWI-ST227:389:C4WA2ACXX:7:1204:2272:59979

Line 2: Sequence

GAGGAAGAGCTCCCTGCCTCTCTCCTCCTTAAAGGAAAGTGGGCGGAAT

Line 3: Seperator Line

+

Line 4: Quality Scores (Phred score conversion)

[illegible]

Each character represents a **score between 0 and ~40+**, indicating the confidence of each base call. High-quality base = high score e.g., 'F' or 'J', Low-quality base = low score e.g., '!' or '#'. ASCII characters encoding the **Phred quality score** of each base in Line 2

Header	Sequence	Quality
@HWI-ST227:389:C4WA2ACXX:7:1204:2272:59979	GGAGGAAGGTCTCTCGTCTCTCTTCATATAAGGGAATGGCTGAAT	+
FFFFHHHHHHJJJJJJJJJJJJJJJJIGIGIGJJJJJJJJJJJJJJ	@HWI-ST227:389:C4WA2ACXX:7:1205:15214:42893	
GAGGATCCCAGGAGGAAGGTCTCTCGTCTCTCTTCATCTAAGGGA	+	
12BAFB?A:3<AE1@<FF;1*(EG*)?0?DBD>9BF9B*?#####	@HWI-ST227:389:C4WA2ACXX:8:2208:2467:44624	
AAAGAGGAGAGAGGACCATCTCCCTGGGATCCTCAGAAGTCTACT	+	
BDDA:DB?2AA@FC>F?EEGC<FED>GFD;?GBB?<?F99*/9?9?		

Phred scores:

F-J: High quality scores (Phred 37-41)

H/G: Very good quality (Phred 33–35)

E-D: Moderate quality (Phred ~30)

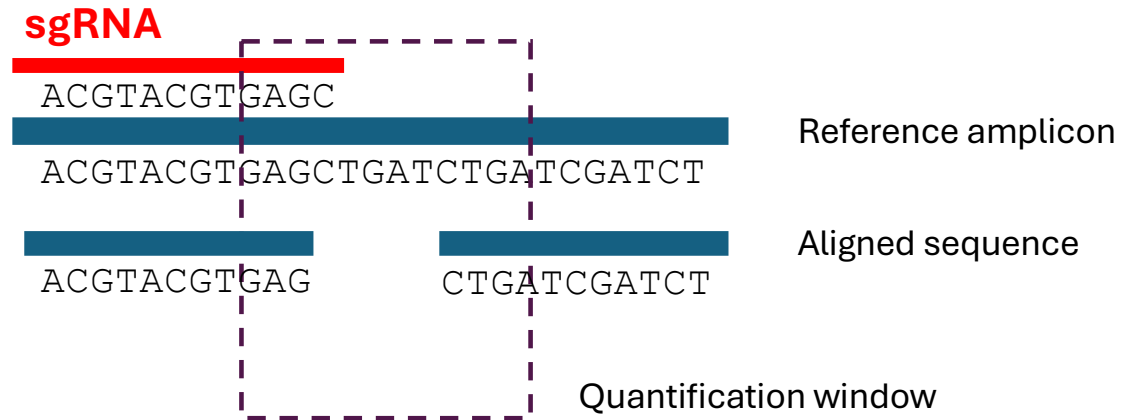
Processing of .fastq files by CRISPResso2

1. Data

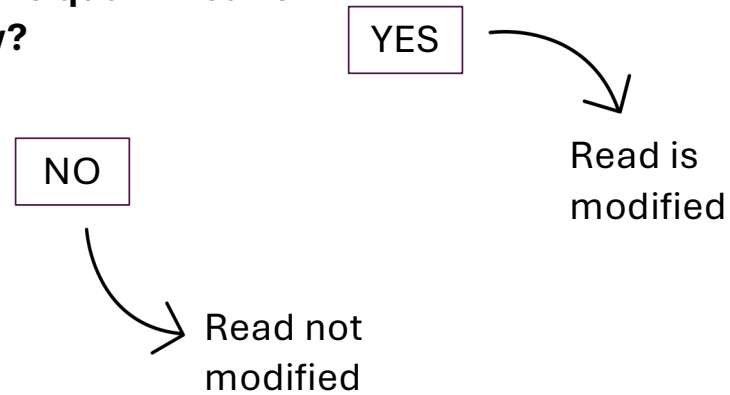
Header	Sequence	Quality
@HWI-ST227:389:C4WA2ACXX:7:1204:2272:59979	GGAGGAAGTCTCTCGTCTCTTTCATATAAGGGAATGCGTGAAT	+
FFFFHHHHHHJJJJJJJJJJJJIGIGIGJJJJJJJJJJII		
@HWI-ST227:389:C4WA2ACXX:7:1205:15214:42893	GAGGATCCCAGGAGGAAGTCTCGTCTCTTTCATCTAAGGGA	+
12BAFBF7A:3<AE1<CF;1*EG*?>0DBD:9BF9B*?#####		
@HWI-ST227:389:C4WA2ACXX:8:2208:2467:44624	AAAGAGGAGAGAGGACCATCTCCCTGGGATCTCAGAAGTCTACT	+
BDDA:DB?2AA<FC>F?EEGC<FED>GFD;?GBB?<?F99*/9?9?		

Sequencing reads (.fastq)

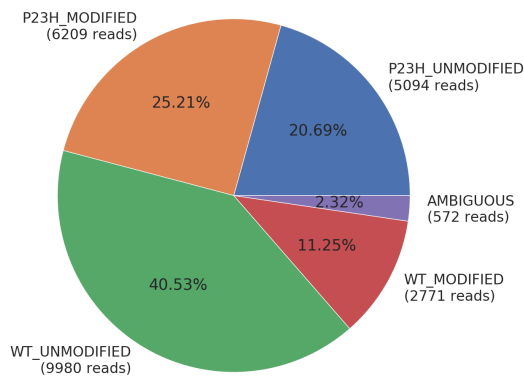
2. Alignment



3. Are edits present within the quantification window?



4. Summary of Editing



Inputs to CRISPResso2

Editing Type

Cas9

Cpf1

Base editors

Prime editors

Custom

Data

1.

Header	Sequence	Quality
0HWI-ST227:389:C4WA2ACXX:7:1204:2272:59979	GGAGGAAGGTCTCGTCTCTTTTCATATAAGGGAATGGCTGAAT	
+		
0HWI-ST227:389:C4WA2ACXX:7:1205:15214:42893	GAGGATCCAGGAGGAAGTCTCGTCTCTTTTCATCTAAGGGA	
+		
12BAFB7A:3<AE18<FF:1*8EG*)707DBD>9BF9B*7#####		
0HWI-ST227:389:C4WA2ACXX:8:2208:2467:44624	AAAGAGGAGAGAGGACCATCTCTCCCTGGGATCTCAGAGTCTACT	
+		
BDDA:DB?2AA8FC>F?EEGC<FED>GFD;?GBB?<?F99*/9?9?		

Fastq File 1

Header	Sequence	Quality
0HWI-ST227:389:C4WA2ACXX:7:1204:2272:59979	GGAGGAAGGTCTCGTCTCTTTTCATATAAGGGAATGGCTGAAT	
+		
0HWI-ST227:389:C4WA2ACXX:7:1205:15214:42893	GAGGATCCAGGAGGAAGTCTCGTCTCTTTTCATCTAAGGGA	
+		
12BAFB7A:3<AE18<FF:1*8EG*)707DBD>9BF9B*7#####		
0HWI-ST227:389:C4WA2ACXX:8:2208:2467:44624	AAAGAGGAGAGAGGACCATCTCTCCCTGGGATCTCAGAGTCTACT	
+		
BDDA:DB?2AA8FC>F?EEGC<FED>GFD;?GBB?<?F99*/9?9?		

Fastq File 2

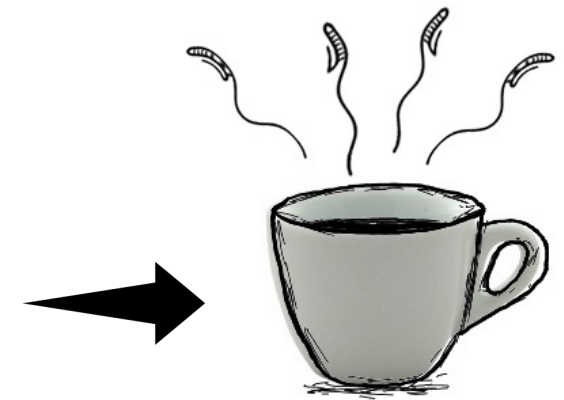
2.

AATGTCCCCCAATGGGAAGTTCATCTGGCACTGCCCACAGGTGAG
GAGGTCATGATCCCCTTCTGGAGCTCCCAACGGGCGGTGGTCTGG
TTCATCATCTGTAAAGAATGGCTTCAAGAGGCTCGGCTGTGGTT

Amplicon

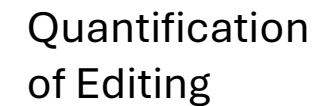
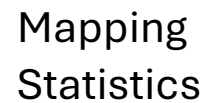
3.

GTGCGGAGCCACTTCGAGCAGC
sgRNA

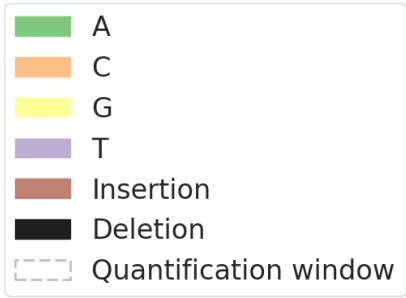
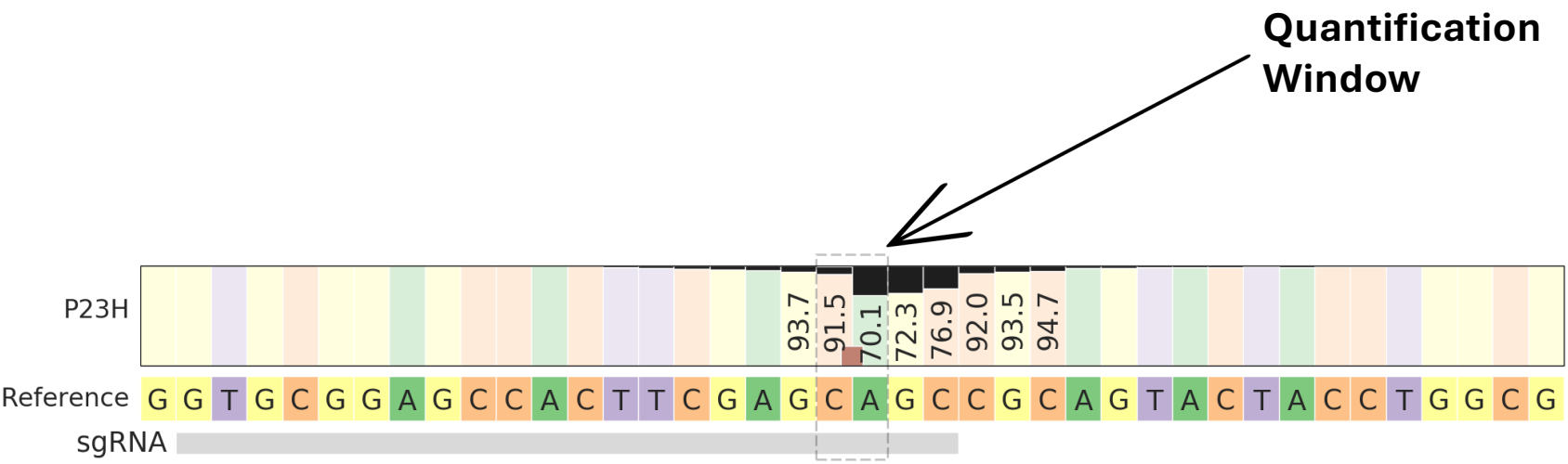


CRISPResso2

CRISPResso2

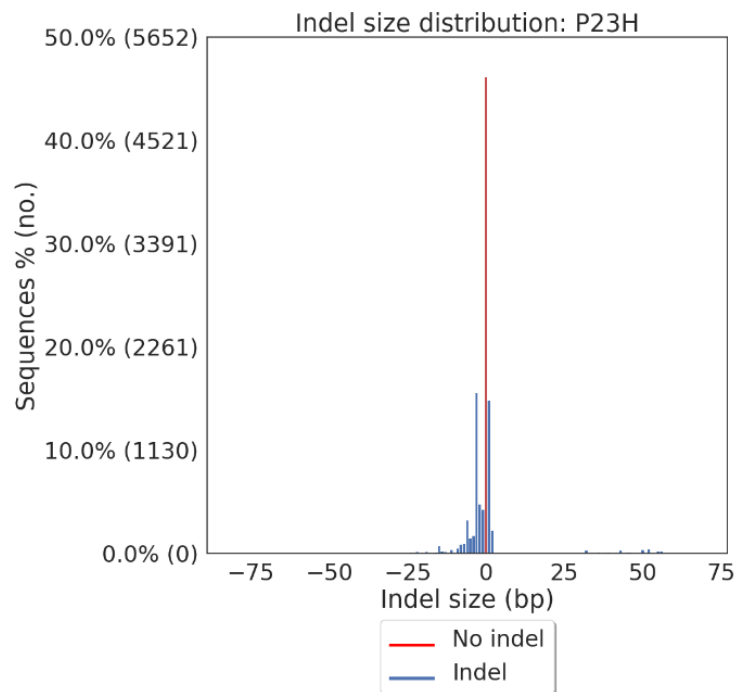


Outputs from CRISPResso2

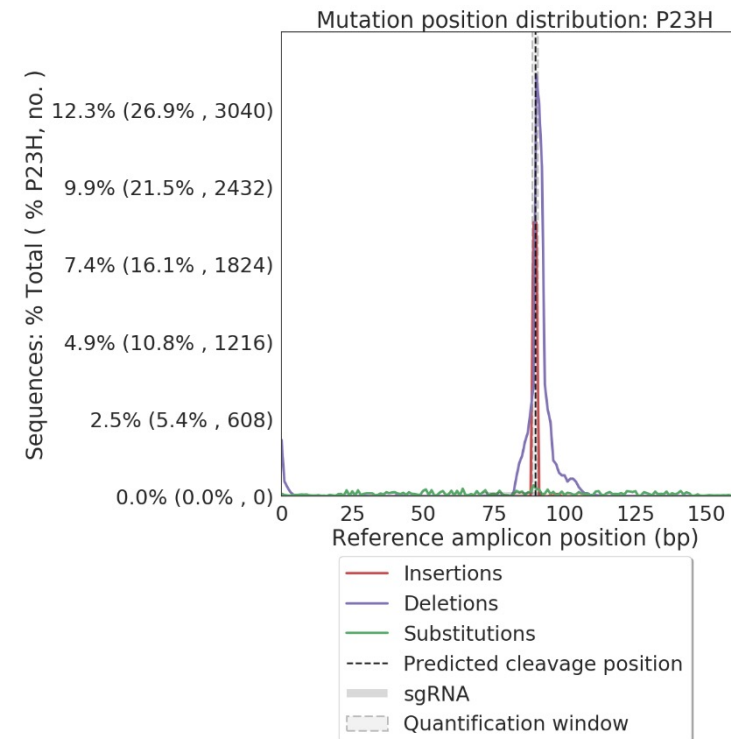


Distribution of nucleotides in the proximity of the sgRNA and cutting site

Outputs from CRISPResso2

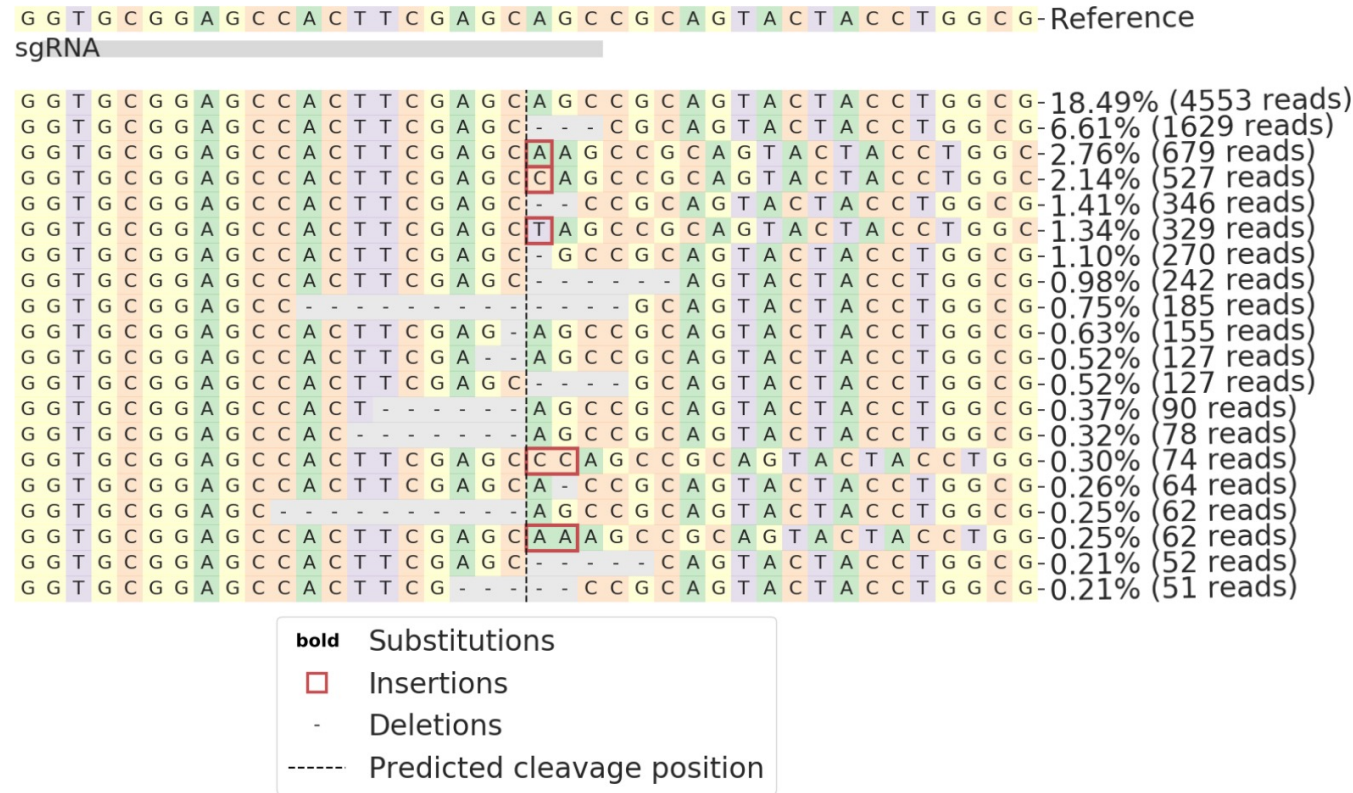


Distribution alleles with and without indels



Frequency of insertions, deletions, and substitutions across the amplicon, including modifications outside of the quantification window.

Outputs from CRISPResso2



Visualization of the distribution of identified alleles around the cleavage site for the sgRNA

GTGCGGAGCCACTTCGAGCAGC. Substitutions are in bold. Red rectangles highlight inserted sequences.

Horizontal dashed lines indicate deleted sequences. The vertical dashed line indicates the predicted cleavage site.

How to use CRISPResso 2 Online (Web)



<https://crispresso2.pinellolab.org/>



CRISPResso2

Analysis of genome editing outcomes
from deep sequencing data

Experimental design

Editing tool:

Cas9 ⓘ Cpfl ⓘ Base editors ⓘ Prime editors ⓘ Custom ⓘ

Sequencing design:

Paired end reads Single end reads Interleaved reads ⓘ

Fastq file R1 Use the Browse button or drag and drop a fastq or f... Browse

Fastq file R2 Use the Browse button or drag and drop a fastq or f... Browse

Amplicon Enter the amplicon sequence. If submitting more than one amplicon, please separate amplicons using commas.

sgRNA ⓘ Enter the sgRNA sequence/s excluding the PAM sequence.

Pros:

- No installation
- Easy to use
- Great interface

Cons:

- Can't use if offline
- Slow if uploading large files
- Slow if many people are using site simultaneously
- Sometimes site is unreachable

How to use CRISPResso2 locally (on own computer)

From official documentation:

<https://docs.crispresso.com/installation.html>

CRISPResso can be installed using the conda package manager Bioconda, or it can be run using the **Docker containerization** system.

Running via docker is probably the easiest way, as you don't have to worry about which packages to install

Docker

CRISPResso can be used via the Docker containerization system. This system allows CRISPResso to run on your system without configuring and installing additional packages. To run CRISPResso, first download and install docker: <https://docs.docker.com/engine/installation/>.

Next, Docker must be configured to access your hard drive and to run with sufficient memory. These parameters can be found in the Docker settings menu. To allow Docker to access your hard drive, select 'Shared Drives' and make sure your drive name is selected. To adjust the memory allocation, select the 'Advanced' tab and allocate at least 4G of memory.

To run CRISPResso, make sure Docker is running, then open a command prompt (Mac) or Powershell (Windows). Change directories to the location where your data is, and run the following command:

```
docker run -v ${PWD}:/DATA -w /DATA -i pinello/CRISPResso2 CRISPResso -h
```



Pros:

- *FAST!!*
- Works offline
- Large files are not an issue
- Docker containers make running it a breeze.
- Platform independent

Cons:

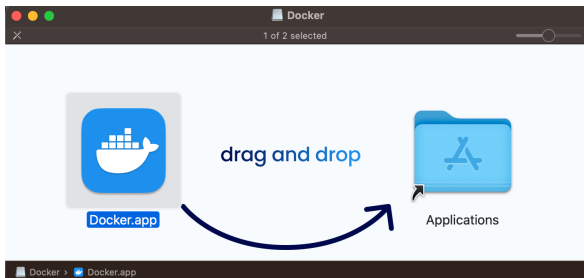
- Need to be slightly more tech savvy
- Command line interface can be daunting for some
- Install Docker
- Some command line knowledge required

Getting Docker up and running (on Apple Mac)

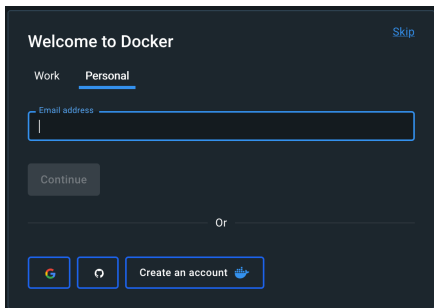
<https://docs.docker.com/desktop/>

<https://docs.docker.com/desktop/install/mac-install/>

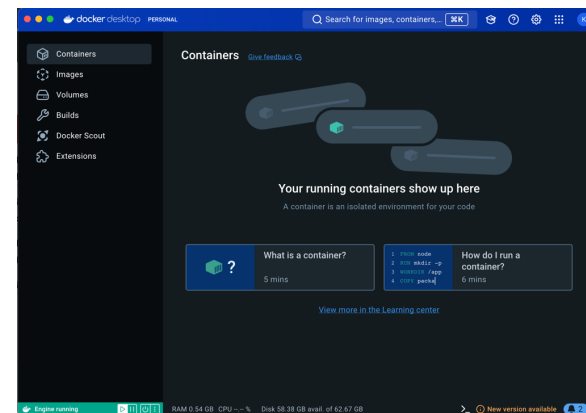
1. Download the installer (Docker.dmg for Mac)
2. Double click then copy to Applications folder



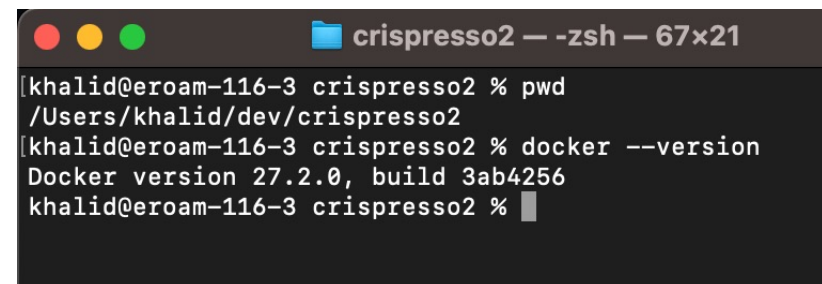
3. Login using email or create a new account



4. Home screen after sign up and logging in



5. Check Docker is running and version via terminal using **docker --version** command.



Run CRISPresso2 via Docker

1. Execute the following command to download latest version of Crispresso2 container

```
docker run -v ${PWD}:/DATA -w /DATA -i pinellolab/crispresso2 CRISPresso -h
```

2. Data to be processed should be copied into working directory (e.g. /dev/crispresso2) – download the files **nhej.r1.fastq.gz** and **nhej.r2.fastq.gz** from their website

3. Run crispresso2 with arguments as follows

```
docker run -v ${PWD}:/DATA -w /DATA -i pinellolab/crispresso2 CRISPresso -r1 nhej.r1.fastq.gz -r2 nhej.r2.fastq.gz -a
```

```
AATGTCCCCCAATGGGAAGTTCATCTGGCACTGCCCACAGGTGAGGAGGTCATGATCCCCTTCTGGAGCTCCCAA  
CGGGCCGTGGTCTGGTTCATCATCTGTAAGAATGGCTTCAAGAGGCTCGGCTGTGGTT
```

4. If executed successfully a **directory** and **HTML** file will be created as follows:

- a) CRISPresso_on_nhej.r1_nhej.r2/
- b) CRISPresso_on_nhej.r1_nhej.r2.html

5. Open HTML file 4(b) in browser to view reports

Task 1 - create a report using the web interface

1. Download data sets from GitHub, start with the pair: **MCF7-L1W_S2_L001_R1_001.fastq.gz** and **MCF7-L1W_S2_L001_R2_001.fastq.gz** – (these are both about 75MB each)

2. Use the amplicon:

```
CCTATGCTTGGACCTAGGTGTCATAACTTACTTTAAATATGTATGTTTGGTTTTTCATTCATATTGACAGTACTACCTCTCAGTTTTCTTTCAGATATT  
GTTTTGTATTTACCCATGAAGACATTGTTTTTTGGACTCTGCAAATAGGACATTTCAAAGATGAGTGAAAAAAATTGGAAACAACTGCACAGCAGCG  
GAAATGTCCTGAATGGATGAATGTGCAGAATAAAAGATGTGCTGTAGAAGAAAGAAAGGTATGTTGTTTCATTGACTATTCTTTTGGGTGAGAAATTTA  
ATTTATATTTGACTGTGCAAAGAGTCAGTTGTTACTTGTAACCTCAAGTCATTGTTTAGGTCAGAGTTGCTGTTGTCTAAATGCACCAGGACCTAGT  
TGTTGAAAGGGTAAACTGGAATAAACTTTAATTGGGTTTACAAAATGAGAATTCTTACTGTATATTTTCTCTTTTTTCGGGTTGACTTTACCAGT
```

3. Use the sgRNA:

```
GGAAACAACTGCACAGCAG
```

4. Use the web interface to upload the **2 guides**, and cut and paste the **amplicon** (above) and the **sgRNA** (above) into the correct fields.

Task 2 - create a report locally using Docker

1. Download data sets from GitHub, start with the pair: **MCF7-L1W_S2_L001_R1_001.fastq.gz** and **MCF7-L1W_S2_L001_R2_001.fastq.gz** – these are both about 75MB each (done in Task 1)

Use this as a *template command* for docker:

```
docker run -v ${PWD}:/DATA -w /DATA -i pinellolab/crispresso2 CRISPResso -r1  
FILENAME1 -r2 FILENAME2 -a AMPLICON -g sgRNAGuide
```

2. Use the amplicon:

```
CCTATGCTTGGACCTAGGTGTCATAACTTACTTTAAATATGTATGTTTGGTTTTTCATTCATATTGACAGTACTACCTCTCAGTTTTCTTTCAGATATTGTTTTGTATTTACCC  
ATGAAGACATTGTTTTTTGGACTCTGCAAAATAGGACATTTCAAAGATGAGTGAAAAAAATTTGGAAACAACTGCACAGCAGCGGAAATGTCCTGAATGGATGAATGTGCAGAA  
TAAAGATGTGCTGTAGAAGAAAGAAAGGTATGTTGTTTCATTGACTATTCTTTTGGGTGAGAAATTTAATTTATATTTGACTGTGCAAGAGTCAGTTGTTACTTGTAACCTT  
CAAGTCATTGTTTAGGTGAGAGTTGCTGTTGTCTAAATGCACCAGGACCTAGTTGTTGAAAGGGTAAACTGGAATAAACTTTAATTGGGTTTACAAAATGAGAATTCCTTACTG  
TATATTTTCTCTTTTTTCGGGTTGACTTTACCAGT
```

3. Use the sgRNA:

GGAAACAACTGCACAGCAG

4. The full command looks like this:

```
docker run -v ${PWD}:/DATA -w /DATA -i pinellolab/crispresso2 CRISPResso -r1 MCF7-L1W_S2_L001_R1_001.fastq.gz -r2 MCF7-  
L1W_S2_L001_R2_001.fastq.gz -a  
CCTATGCTTGGACCTAGGTGTCATAACTTACTTTAAATATGTATGTTTGGTTTTTCATTCATATTGACAGTACTACCTCTCAGTTTTCTTTCAGATATTGTTTTGTATTTACCCATGAAGACATTGTTTTTTGGACTC  
TGCAAAATAGGACATTTCAAAGATGAGTGAAAAAAATTTGGAAACAACTGCACAGCAGCGGAAATGTCCTGAATGGATGAATGTGCAGAAATAAGATGTGCTGTAGAAGAAAGAAAGGTATGTTGTTTCATTGACTAT  
TCTTTTGGGTGAGAAATTTAATTTATATTTGACTGTGCAAGAGTCAGTTGTTACTTGTAACCTTCAAGTCATTGTTTAGGTGAGAGTTGCTGTTGTCTAAATGCACCAGGACCTAGTTGTTGAAAGGGTAAACTGG  
AATAAACTTTAATTGGGTTTACAAAATGAGAATTCCTTACTGTATATTTTCTCTTTTTTCGGGTTGACTTTACCAGT -g GGAAACAACTGCACAGCAG
```

Sample command line output

```
khalid@eroam-116-3 crispresso2 % docker run -v ${PWD}:/DATA -w /DATA -i pinellolab/crispresso2 CRISPResso -r1 MCF7-L1W_S2_L001_R1_001.fastq.gz -r2 MCF7-L1W_S2_L001_R2_001.fastq.gz -a CCTATGCTTGGACCTAGGTGTCATAACTTACTTTAAATATGTATGTTTGGTTTCATTTCATTTGACAGTACTACCTCTCAGTTTTCTTTCAGATATTGTTTGTATTTACCCATGAAGACATTGTTTTTGGACTCTGCAAAATAGGACATTTCAAAGATGAGTGAAAAAAATTGGAACAACCTGCACAGCAGCGGAAATGCCTGAATGGATGAATGTCAGAAATAAAGATGTGCTGTAGAAGAAAGAAAGGTATGTTGTTTCATTGACTATTCTTTTGGGTGAGAAATTTAATTTATATTTGACTGTGCAAGAGTCAGTTGTTACTTGTAAACTTCAAGTCATTGTTTAGGTCAGAGTTGCTGTTGTCTAAATGCACCAGGACCTAGTTGTTGAAAGGGTAACTGGAATAAACTTTAATTGGGTTTACAAAATGAGAATTCTTACTGTATATTTTCTCTTTTTCGGGTTGACTTTACCAGT -g GGAAACAACCTGCACAGCAG
WARNING: The requested image's platform (linux/amd64) does not match the detected host platform (linux/arm64/v8) and no specific platform was requested
INFO @ Mon, 07 Oct 2024 18:30:25:
    Creating Folder CRISPResso_on_MCF7-L1W_S2_L001_R1_001_MCF7-L1W_S2_L001_R2_001

INFO @ Mon, 07 Oct 2024 18:30:25:
    Computing quantification windows

INFO @ Mon, 07 Oct 2024 18:30:25:
    CRISPRessoPro not installed

INFO @ Mon, 07 Oct 2024 18:30:25:
    Added 0 guides with flexible matching
    Original flexiguides: ['None']
    Found guides: []
    Mismatch locations: []

INFO @ Mon, 07 Oct 2024 18:30:26:
    Processing sequences with fastp...

INFO @ Mon, 07 Oct 2024 18:30:39:
    Done!

INFO @ Mon, 07 Oct 2024 18:30:39:
    Done!

INFO @ Mon, 07 Oct 2024 18:30:40:
    Aligning sequences...

INFO @ Mon, 07 Oct 2024 18:30:40:
    Processing reads; N_TOT_READS: 0 N_COMPUTED_ALN: 0 N_CACHED_ALN: 0 N_COMPUTED_NOTALN: 0 N_CACHED_NOTALN: 0

INFO @ Mon, 07 Oct 2024 18:30:49:
    Processing reads; N_TOT_READS: 10000 N_COMPUTED_ALN: 841 N_CACHED_ALN: 6731 N_COMPUTED_NOTALN: 447 N_CACHED_NOTALN: 1981

INFO @ Mon, 07 Oct 2024 18:30:57:
    Processing reads; N_TOT_READS: 20000 N_COMPUTED_ALN: 1474 N_CACHED_ALN: 13479 N_COMPUTED_NOTALN: 845 N_CACHED_NOTALN: 4202

INFO @ Mon, 07 Oct 2024 18:31:04:
    Processing reads; N_TOT_READS: 30000 N_COMPUTED_ALN: 2023 N_CACHED_ALN: 20265 N_COMPUTED_NOTALN: 1189 N_CACHED_NOTALN: 6523

INFO @ Mon, 07 Oct 2024 18:31:10:
    Processing reads; N_TOT_READS: 40000 N_COMPUTED_ALN: 2525 N_CACHED_ALN: 27023 N_COMPUTED_NOTALN: 1534 N_CACHED_NOTALN: 8918
```

Post analysis steps

- Interpret allele frequencies and mutation types.

Assess Editing Efficiency: Examine the percentage of edited alleles compared to the wild-type sequence. Low editing efficiency might indicate issues with sgRNA design, Cas9 delivery, or cell type.

Mutation Types: Determine the predominant mutation types (insertions, deletions, substitutions) and whether they match the expected outcome (e.g., frameshift mutations, knockout efficiency).

On-target and Off-target Analysis: Review any off-target effects or mutations in other regions if genome-wide CRISPR sequencing was performed.

- Validate your results with sequencing and functional assays.

Functional Assays: If the study involves gene knockout or specific genetic modifications, functional assays can determine if the edited cells behave according to the hypothesis (e.g., loss of a phenotype, change in metabolic activity, or altered growth patterns).

- Troubleshoot any inefficiencies (redesign sgRNA, optimize delivery).

Low Editing Efficiency: If CRISPResso2 shows a low rate of edits, troubleshoot by: redesigning the guide RNA (sgRNA) for better targeting. Optimizing the transfection or delivery method of Cas9 and the guide RNA (e.g., improving Cas9 or sgRNA concentration, delivery method). Ensuring that the Cas9 variant being used matches the goal (wild-type Cas9 for cutting, Cas9 nickase for precise editing, etc.).

- Isolate specific clones if necessary.

Clone Isolation: If aiming for a homogeneous population of cells with specific mutations (especially for knockout studies or disease modelling), proceed with **clonal isolation**.

- Document and report results.

Acknowledgements

Developers of CRISPResso2

Luca Pinello Ph.D., Associate Professor
Kendell Clement, Assistant Professor

Genome Engineering Laboratory (University of Westminster)

Dr Kalpana Surendranath, Associate Professor, University of Westminster
Dr John Murphy, Professor, University of Westminster
Dr Munuse Savash, Ph.D, University of Oxford
Dr Varsha Tieote, Ph.D, University of Westminster
Dr Nadine Solaiman, Ph.D, University of Westminster
Dr Ahmed Sidali, Ph.D, University of Westminster

All the M.Sc. and B.Sc. students who have contributed so much to the laboratory and the Gene Editors of the Future program.

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(<https://quintinhoggtrust.org/>)

Gene Editors of The Future (University of Westminster)

Big thank you to all the students from iterations 1-4 of this program

Any questions?



Contact

Email: **k.akram@westminster.ac.uk**

LinkedIn: <https://www.linkedin.com/in/khalidakram/>

UNIVERSITY OF
WESTMINSTER



wellcome
connecting
science

